

Max Zhu

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OBJECTIVE

Motivated Computer Engineering student strongly interested in computer architecture and FPGA. Seeking a Summer 2025 internship to gain real world experience in digital design, hardware-software integration, and low-level programming. Eager to contribute to innovative projects, determined to expand my knowledge, and excited to collaborate with a like-minded team to solve complex challenges in a dynamic environment.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, GA

B.S. in Computer Engineering

Expected Graduation May 2026

GPA: 3.82/4.0

Major GPA 4.0/4.0

- **Threads:** Computing Hardware and Emerging Architecture and Systems and Architecture
- **Relevant Coursework:** Digital System Design, Circuit Analysis, HW/SW Programming, Data Structures and Algorithms, Object Oriented Programming, Linear Algebra, Differential Equations

EXPERIENCE

Georgia Tech School of Electrical and Computer Engineering

Atlanta, GA

Opportunity Research Scholar for Wireless Twin Digital Systems

August 2024 – Present

- Analyzing research papers on 5G, Wi-Fi 6, ray tracing, wireless signals, and UAVs to optimize and develop twin digital systems
- Implementing Sionna, Blender, Wireless Insight, and radio environments to enhance the creation of wireless twin digital systems

Knack

Atlanta, GA

Student Tutor

August 2024 – Present

- Created tailored learning plans to teach students complex concepts while maintaining engagement
- Taught courses including Digital System Design, Circuit Analysis, Differential Equations, DSA, and OOP

Metro Healthcare

Vancouver, BC

IT Specialist

November 2023 – January 2024

- Integrated digital infrastructure like POS systems, shared drives, website access, and patient database across multiple locations
- Developed a clinical website to showcase services and streamline appointment scheduling

PROJECTS

Cloud Server

July 2024 – Present

- Configured and deployed a cloud server with Ubuntu Server to replace multiple personal subscription services
- Exclusively utilized Linux command line for system administration and server organization

Mbed Wordle Game

September 2024 – November 2024

- Created a Wordle-like game using C++, NAV Switch, Mbed, LPC1768, and an LCD for an interactive interface
- Used Google Tests to ensure correct back-end functionality and ensured no memory leaks

Spotify Car Thing

September 2024 – November 2024

- Developed a desktop device inspired by the “Spotify Car Thing” allowing for live interactions with the Spotify interface
- Integrated physical inputs and utilized the ESP8266, C++, and Spotify API to skip, like, pause, and view the current

Electronics Store

September 2020 – August 2023

- Sourced, assembled, and sold custom desktops, cables, and keyboards, achieving significant sales growth
- Traced printed circuit boards and replaced components to repair a wide range of electronics including phones and laptops

LEADERSHIP AND EXTRACURRICULARS

SILICON JACKETS

Atlanta, GA

Digital Design Member

August 2024 – Present

- Designed and implemented CPU functionality using Verilog, focusing on state machines for optimized performance
- Developed and executed comprehensive test benches to verify hardware designs, ensuring accuracy and reliability

HIVE

Atlanta, GA

Peer Instructor

December 2023 – Present

- Taught and guided end-users on laser cutters, 3d printers, shop machines, and benchtop tools
- Led interactive workshops on circuit design, skill implementation, and product design

SKILLS

Software: C, C++, VHDL, Java, Python, Verilog, Git, CUDA, Ubuntu, MatLab, Sionna, Quartus, HTML, Javascript

Hardware: FPGA, Oscilloscopes, Digital Circuits, PCB Design, Arduino